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Roll No

EC-704-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Microwave Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Prove with the help of Maxwell's equations that TEM mode cannot exist in rectangular waveguide.
2. Consider a circular wave guide with $a = 0.8\text{cm}$. Compute the cut-off frequencies and identify the first four propagating modes.
3. Obtain scattering matrix of magic tee.
4. Write short notes on the following:
 - a) E-plane TEE
 - b) Directional coupler
5. What are avalanche transit time devices? Explain the operation, construction and applications of IMPATT.

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6. a) Explain the square law detection. Discuss the principle of broadband and tuned detectors.
b) Explain TRAPATT and BARITT.
7. Draw the schematic diagram of TWT amplifier and describe its principle of operation.
8. Write short notes on the following :
 - i) Network analyzer
 - ii) Measurement of scattering matrix parameters.
